Finite-Support Periodic Highways of Langton's Ant

Research Report Sample

Project researcher: Atharva Jillhewar

Research evidence through 12 October 2026
Sample revised 12 October 2026

Abstract

A repeating turn word defines a highway of Langton's ant only if visits from successive copies request compatible colours at each physical cell. A necessary and sufficient finite test checks this compatibility and constructs a finite black seed for every word that passes. Starting from the least seed, each period-boundary state consists of the translated seed and disjoint layers of accumulated wake. For a heading-resetting word of length N and nonzero displacement d, stable cell chronology identifies the black-count increase g with the number of odd translation classes. Conservation then gives $g \geq 2\|d\|_\infty$. Analytic drift exclusions and a symbolic computer-assisted primitive-diagonal argument, with no period cutoff, imply $g \in 4\mathbb{Z}_{\geq 2}$. For diagonal drift, geometric rigidity and finite strip certificates force transverse width $W \geq 10$. An explicit period-104, growth-12 word with an eleven-cell seed attains that width. First-entry cycles and integer detour charges yield $N \geq 6\|d\|_\infty + 2\|d\|_1$, sharpened to $N \geq 10m + 4$ for $d = (m, -m)$, $m > 0$. Routing and incidence identities, polynomial charges and black lifetimes give further constraints. Formal count walks still require physical colour consistency. Archived exhaustive searches exclude all nonzero-drift periods through 48 and growth-eight periods through 60. The computational premises are stated explicitly. All ten growth-eight drift orbits retain open cases, and eventual highway entry from every finite seed remains unproved.

1 Research question and exact framework

Langton's ant turns right on a white cell and left on a black cell. It then flips the cell and moves one unit in its new heading [7]. The initial black set is finite. The familiar all-white trajectory develops a period-104 highway, but universal entrance into that highway remains open. Unboundedness [1], checkerboard routing [6], universality on suitable infinite backgrounds [4], and symbolic trace descriptions [5] do not settle the finite-seed question. Related highway and confinement work supplies context [3, 8, 2].

Which finite turn words can repeat indefinitely from a finite black seed, and what restrictions follow on growth, drift, width and period? Copies of a word can revisit a cell in a different phase order. The criterion below checks their colour requirements in physical order, then uses that stable chronology to derive the restrictions. Unboundedness and checkerboard routing are established results. The criterion and its consequences are developed here, with proof sources and executable evidence identified in the accompanying archive [10].

Write a state as (B, p, q), with finite $B \subset \mathbb{Z}^2$, position p, and heading $q \in \mathbb{Z}/4\mathbb{Z}$, with $0 = N, 1 = E, 2 = S, 3 = W$ and unit vectors $(u_0, u_1, u_2, u_3) = ((0,1), (1,0), (0,-1), (-1,0))$. A step with turn sign $r = 1$ on white and $r = -1$ on black is

$$(B, p, q) \longmapsto (B \triangle \{p\}, p + u_{q+r}, q + r).$$

A word $w_0 \cdots w_{N-1}$ induces positions $p_0, \ldots, p_N$. Require heading reset and nonzero drift $d = p_N - p_0 = (a, b)$, and put $g = \#R - \#L$. Each right turn adds one black cell and each left turn removes one, so g is the black-count increase per period. The period need not be minimal. Translation and cyclic phase changes are allowed, as are quarter-turn rotations. Reflection changes chirality and is excluded. A periodic highway is a finite-seed orbit whose turn trace eventually repeats such a heading-resetting word with nonzero drift.

Principal results and proof route. Theorem 2.1 decides a proposed word and constructs its seed. Wake chronology gives the growth–drift bound (Theorem 3.2). Geometry and strip certificates give diagonal width $W \geq 10$ (Theorem 5.3), attained by the explicit example. This width bound also supports the primitive growth-four exclusion; together the exclusions in Section 6 give $g \geq 8$. Selected cycles bound period (Theorem 7.1), sharpened to $N \geq 10m + 4$ for drift $(m, -m)$. Other identities give necessary conditions. The exhaustive searches exclude only the finite ranges stated.

Lemma 1.1 (Elementary invariants)**.** *The quantities* $q_t - |B_t| \pmod 4$ *and* $q_t - (p_{t,x} + p_{t,y}) \pmod 2$ *are constant. At each physical cell the arrival axis is fixed and the turns alternate.*

Checkerboard routing is classical [6]; the proof fixes this report's conventions.

Proof. Heading and black count both change by r_t modulo four. Heading parity and position checker parity both flip each step. Thus every visit to a fixed cell has the same heading parity, giving its fixed horizontal or vertical arrival axis. Flipping that cell after each visit forces its successive colours, and therefore turns, to alternate. □

Heading reset gives $g \equiv 0 \pmod 4$, hence even N and even $a + b$. The $N/2$ horizontal and $N/2$ vertical steps also give $a \equiv b \equiv N/2 \pmod 2$.

The update is reversible: from the new pose, step backward in its current heading, flip that previous cell, and undo the uniquely determined turn. A clean translator satisfies $B_N = B_0 + d$, with position translated by d and heading reset. Reversibility extends this whole-state translation to negative times. It gives no periodicity conclusion for a general finite seed.

Unboundedness has a geometric proof [1]. Let S_∞ be the infinitely visited cells. Two adjacent neighbours of a recurrent cell cannot both lie outside S_∞: after their last visits, only one arrival arc remains, but alternating turns require both outgoing arcs. A bounded orbit would have finite nonempty S_∞; its $x + y$-maximizer violates this condition. In a finite-height recurrent strip, each top-level cell similarly forces both horizontal neighbours recurrent, hence a bi-infinite row. Ruling out such rows requires more than unboundedness.

2 A constructive finite test for periodic realisability

Group the phases by $i \sim j$ when $p_i - p_j \in \mathbb{Z}d$. For a class I, choose b_I and integers a_i with $p_i = b_I + a_i d$. At the physical cell $b_I + nd$, phase i has visit time $i + N(n - a_i)$ when $n \geq a_i$. Order those visits by time and call the resulting symbol sequence $S_{I,n}$.

Theorem 2.1 (Exact criterion)**.** *The repeated word w^ω is realisable from a finite black seed if and only if every $S_{I,n}$ alternates and each stabilised class word begins with R. It suffices to check* $\min_{i \in I} a_i \leq n \leq \max_{i \in I} a_i$. *A seed is*

$$B = \{b_I + nd : \min a_i \leq n < \max a_i,\ S_{I,n} \text{ begins with } L\}.$$

Proof. Phase i in period $k \geq 0$ visits $b_I + nd$ exactly when $k = n - a_i$, giving the stated chronology. Cell flips force alternation. For $n \geq \max a_i$, the order is independent of n; finite support forces this common first turn to be R, since infinitely many such cells are white. Conversely, the displayed seed assigns each boundary cell its required first colour. All stabilised cells start white; alternation then supplies every later requested colour. Induction on time reproduces the proposed word, with heading reset and translation by d after each period. □

Sort by decreasing a_i, breaking ties by phase. Every boundary word is a suffix of the stabilised word and inherits its alternation.

Corollary 2.2 (Least seed for a fixed trace)**.** *Fix an accepted trace starting at time zero, and let $V = \bigcup_i (p_i + \mathbb{Z}_{\geq 0} d)$. Every finite realising seed is $B \sqcup F$ with finite $F \subseteq \mathbb{Z}^2 \setminus V$. Consequently B is the unique least seed. If a class has ordered levels $a_1 \geq \cdots \geq a_m$, its seed contribution is $\sum_{1 \leq j < m,\ j \text{ odd}} (a_j - a_{j+1})$.*

Proof. The first turn at every visited cell forces its initial colour; cells outside V are never read. The active suffix starts L exactly on the integer intervals $[a_2, a_1), [a_4, a_3), \ldots$; tied levels give empty intervals. □

Example: cross-period chronology. For the realised word of Section 9, take $N = 104$, $d = (2, -2)$ and the class representative $b = (1, -2)$. Its phase/level pairs in stabilised chronological order are

$$(53, 1), (29, 0), (33, 0), (37, 0), (101, 0), (5, -1), (17, -1), (97, -1).$$

The order is by $\tau_i = i - 104 a_i$; phase order alone is wrong. At level $n = -1$ only the last three events are present, giving LRL; at $n = 0$ the word is LRLRLRL; at $n \geq 1$ it is RLRLRLRL. Consequently this class

contributes the two seed cells $b - d = (-1, 0)$ and $b = (1, -2)$. Both occur in the eleven-cell seed displayed with the full word. For exact class indexing write $d = h(a', b')$, where $h = \gcd(|a|, |b|)$ and $\gcd(a', b') = 1$. Choose r, s with $ra' + sb' = 1$. The complete coset key is $\big(b'x - a'y,\ (rx + sy) \bmod h\big)$. The coordinate matrix has determinant $b's + a'r = 1$, and translation by d changes its coordinates by $(0, h)$. This proves completeness. The simpler key $(b'x - a'y, (a'x + b'y) \bmod h)$ fails: for $d = (2, 2)$ it merges $(0, 0)$ and $(1, 1)$, whose difference is not a multiple of d.

3 Wake decomposition, charges and the growth–drift bound

Let $\chi(x, y) = (-1)^{x+y}$. A stabilised class word is $RLRL\cdots$, so an even class contributes zero to g and an odd class contributes one. If O is the set of odd classes, then $g = |O|$. The odd classes form the permanent wake. Define the finitely supported integer field $D(z) = \#\{i : p_i = z,\ w_i = R\} - \#\{i : p_i = z,\ w_i = L\}$ over one period, and write $\delta_c(z) = \mathbf{1}_{z=c}$ for the unit point mass. This signed field is distinct from the colour indicator modulo two.

Proposition 3.1 (Canonical endpoint decomposition)**.** *Let B be the least seed. For each odd class let $c_o = p_i$ at its last stabilised event, and put $W = \{c_o : o \in O\}$. For every $n \geq 0$,*

$$B_{nN} = (B + nd) \,\dot\cup \bigsqcup_{0 \leq k < n} (W + kd), \qquad D = \sum_{c \in W} \delta_c + (T_d - 1)\mathbf{1}_B,$$

where $(T_d A)(z) = A(z - d)$. For arbitrary odd-class representatives, $D = S + (T_d - 1)A$ still holds with $S = \sum_{o \in O} \delta_{c_o}$ and a finite integer-valued A.

Proof. After one period the future path is $V + d$, with least seed $B + d$. In each class, $V \setminus (V + d)$ is its lowest visited cell. Its last stabilised turn is R for an odd class and L for an even class; thus the black cells left outside the future path are exactly W. They are never read again. Iteration proves the union formula: the kth wake layer is below the class minima of the future seed for $k < n$. Signed toggles give $D = \mathbf{1}_{B_N} - \mathbf{1}_B$. Moving a point mass to another representative changes it by $(T_d - 1)$ applied to a finite signed interval, proving the arbitrary-base version. Here W may intersect B; it is $B + d$ that is disjoint from W. □

Let μ_s be the least seed size for cyclic phase s, and ζ_s the indicator of a terminal odd-class event. Advancing the cut discards only its current cell at a terminal R, so $\mu_{s+1} = \mu_s + r_s - \zeta_s$ and $\mu_N = \mu_0$. At a minimum-seed phase, every turn-prefix sum is the nonnegative seed-size change plus $\sum \zeta_s$; its first turn is R because L lowers μ. Thus least-seed and nonnegative-prefix normalisation coexist.

Pairing the decomposition with a drift-periodic conserved weight cancels the endpoint correction and constrains the permanent wake. An additive colour charge has the form $\sum_{z \in B} w(z) + F_q(p) \pmod 4$. It is conserved if

$$F_{q+1}(z + u_{q+1}) - F_q(z) = -w(z), \qquad F_{q-1}(z + u_{q-1}) - F_q(z) = w(z).$$

Eliminating the potentials gives the zero-four-corner equation

$$w(x, y) + w(x + 1, y) + w(x, y + 1) + w(x + 1, y + 1) = 0.$$

Modulo four the converse construction below proves sufficiency. Since $v = \chi w$ has zero mixed difference, the complete weight family is $w(x, y) = \chi(x, y)(A(x) + B(y))$.

No polynomial assumption is needed for the converse construction. The local step equations first force $F_S = F_N + 2w = F_N - 2w$, hence $4w = 0$ in the coefficient group. Thus the corner equation alone is insufficient for integer-valued weights.

Modulo four, prescribe the following diagonal increments:

$$F_N(x+1, y+1) - F_N(x, y) = -w(x, y) + w(x+1, y),$$
$$F_N(x+1, y-1) - F_N(x, y) = w(x, y-1) - w(x+1, y-1).$$

The circulation on each elementary diamond is the negative sum of the two adjacent four-corner expressions, up to a multiple of four. It vanishes. The two diagonal steps generate each checkerboard sublattice, so integration from a basepoint defines F_N independently of path. Define F_E, F_W from the N-heading step equations and put $F_S = F_N - 2w$. Substitution verifies the remaining step equations; the only ambiguity is $4w = 0$. This proves the stated sufficiency. Over $\mathbb{F}_2$ the checker sign disappears and the complete additive weight family is $A(x) + B(y)$. Both classifications leave nonlinear and chronology-dependent invariants open.

Theorem 3.2 (Signed wake residues and density)**.** *For $a \neq 0$, every residue $j \bmod |a|$ satisfies*

$$\sum_{o:\, c_{o,x} \equiv j \ (|a|)} \chi(c_o) \equiv 2 \pmod 4.$$

The analogous statement holds for $b \neq 0$ and $y \bmod |b|$. A zero coordinate instead gives signed sum zero modulo four on every exact-coordinate fibre. Consequently $g \geq 2\max(|a|, |b|) > 0$.

Proof. Take any $\alpha : \mathbb{Z}/|a|\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ and $w(x, y) = \chi(x, y)\alpha(x)$. Define f by $f(x+1) - f(x) = -\alpha(x) - \alpha(x+1)$ and set

$$F_N = \chi f, \qquad F_E = \chi(-f - \alpha),$$
$$F_S = \chi(f - 2\alpha), \quad F_W = \chi(-f + \alpha).$$

Substitution verifies all eight local conservation equations. Across one x-period, $f(x + |a|) - f(x) = -2\sum_j \alpha(j)$. Since $a + b$ is even, translation by d preserves χ; all four potentials have monodromy $2\sum_j \alpha(j) \pmod 4$, for either sign of a. The weight is d-periodic, so the $(T_d - 1)A$ part of Proposition 3.1 cancels in its pairing with w. Conservation over a period becomes

$$\sum_o \chi(c_o)\alpha(c_{o,x}) + 2\sum_j \alpha(j) \equiv 0 \pmod 4.$$

Taking α to be a single-residue indicator proves the x formula. Quarter-turn rotation gives the y formula. For $a = 0$, use an exact-column indicator; b is even and the potential monodromy vanishes.

A sum of signs congruent to two modulo four requires at least two signs. When the coordinates are nonzero, there are $|a|$ fibres in one direction and $|b|$ in the other, giving the bound. Since $d \neq 0$, at least one of these fibre families exists, which also excludes $g = 0$. Together with Lemma 1.1, this gives $g \in 4\mathbb{Z}_{>0}$. □

An exactly two-element fibre must have equal checker signs. At $d = (2, -2)$ and $g = 4$, both coordinate residues have size two, forcing the residue-count matrix to be either the doubled diagonal or doubled off-diagonal. This equality case is used in the growth-four proof.

Let A count odd classes with vertical arrivals and write $N = 2M$. Pair each outgoing horizontal sign $u_j = \pm 1$ with its preceding vertical sign $v_j = \pm 1$. Their sums are a, b; stable alternation gives $\sum_j u_j v_j = A$. Since each product below is zero or four,

$$M + a + b + A = \sum_{j=1}^{M} (1 + u_j)(1 + v_j) \equiv 0 \pmod 4.$$

4 An independent clean-translator obstruction

A clean finite translator repeats its whole finite black set translated: $B_N = B_0 + d$, $p_N = p_0 + d$, $q_N = q_0$. It emits no wake. Over $\mathbb{F}_2$, the Laurent quantities

$$\sum_{(x,y)\in B_t} X^x + V_t X^{p_{t,x}}, \qquad \sum_{(x,y)\in B_t} Y^y + H_t Y^{p_{t,y}}$$

are conserved, where V_t, H_t indicate the arrival axis. A toggle cancels the disappearing or appearing axis correction; the perpendicular move preserves the required coordinate. If both drift coordinates were nonzero, translation would force both polynomials to vanish in an integral domain. Their evaluations at one instead sum to $V_t + H_t = 1$. Thus every clean translator is axis-aligned.

Theorem 4.1 (Even winding)**.** *There is no nonzero-drift clean finite translator.*

Proof. Rotate its drift to $(L, 0)$ with $L > 0$ even. Reversibility extends the translated orbit to all integer periods. Quotient by $(L, 0)$ to obtain a closed cylinder walk of winding one. At a cylinder vertex v, its visits correspond to all visits at one physical representative over the bi-infinite trace. That cell is white at sufficiently early and late times because the black support translates. Hence its visits alternate from R to L: there are k_v of each.

Let H_-, H_+, V_-, V_+ be its adjacent edge multiplicities. The fixed arrival axis gives $H_- + H_+ = V_- + V_+ = 2k_v$. Horizontal multiplicity parity is therefore constant along a row; denote it by h_y. Vertical parity is constant up a column and vanishes outside the finite visited height, so all vertical multiplicities are even. Write them as $2G$.

At a vertical-arrival vertex, let a_R, a_L count arrivals moving north and b_R, b_L arrivals moving south. Then $H_+ = a_R + b_L$, $V_+ = b_R + b_L$, and $k_v = a_R + b_R$. Thus $H_+ - k_v \equiv V_+ \equiv 0 \pmod 2$, giving $k_v \equiv h_y$. At a horizontal-arrival vertex, let a_R, a_L count arrivals moving east and b_R, b_L those moving west. Then $H_+ = b_R + b_L$ and $V_+ = a_L + b_R$, so $H_+ + V_+ \equiv a_L + b_L = k_v \pmod 2$. This gives the same relation. Opposite geometric edges remain distinct when $L = 2$, even though the quotient has parallel edges. For fixed x,

$$\sum_y h_y \equiv \sum_y k_{x,y} = \sum_y (G_{x,y-1/2} + G_{x,y+1/2}) \equiv 0 \pmod 2.$$

Every vertical half-multiplicity occurs twice. The winding parity is the sum of horizontal seam multiplicities, namely $\sum_y h_y$, so winding is even. The displacement $(L, 0)$ requires winding one, a contradiction. □

If $g = 0$, Proposition 3.1 gives $B_N = B + d$, contradicting the clean-translator obstruction without using the density theorem.

5 Diagonal rigidity and the sharp width bound

For diagonal drift normalise $d = (m, -m)$ with $m > 0$, set $t = x + y$, and define $W = t_{\max} - t_{\min}$.

Theorem 5.1 (Extremal-line rigidity)**.** *Both extreme levels have horizontal arrivals. Every visited translation class there consists of a single R event, and W is even.*

Proof. At the maximum level an arrival raises t, while departure must lower it. A vertical arrival would move north; its only permitted turn is L, contradicting the stable first-R condition. A horizontal arrival moves east and must turn R to move south. At the minimum, a vertical arrival moving south likewise requires L; a horizontal arrival moving west turns R to move north. Thus all stable symbols at either extreme are R. Alternation allows only one. The two horizontal levels have the same checker parity, so their difference is even. □

Only visited classes are singletons; some of the m lanes may stay empty.

Proposition 5.2 (Diagonal line laws)**.** *Let m_t count period visits at level t and ω_t its odd classes. Then $m_t \equiv \omega_t \pmod 4$ on horizontal levels and $m_t \equiv -\omega_t \pmod 4$ on vertical levels. If horizontal levels are indexed odd and f_t counts upward cut crossings, then*

$$f_t \equiv (-1)^{t+1} \sum_{s \le t} \omega_s \pmod 4.$$

Horizontal and vertical wake counts α, β satisfy $\alpha \equiv \beta \equiv m \pmod 2$.

Proof. For the closed transverse walk, local maxima and minima have difference $f_{t-1} - f_t$, while $m_t = f_{t-1} + f_t$. Thus the number of local reversals is m_t modulo two. R reverses t on a horizontal level and L reverses it on a vertical level. The complementary turn count is even. A stable class of size $2q + e$ contributes q L turns and $q + e$ R turns. Summing over a level gives the two mod-four laws. The cut formula follows by induction from the empty cut below the trace. There are $N/2$ horizontal steps of net displacement m, so $N \equiv 2m \pmod 4$. Summing level laws gives $N \equiv \alpha - \beta \pmod 4$; together with $\alpha + \beta = g \equiv 0 \pmod 4$ this gives the claimed parity. □

There is a complementary longitudinal law. Put $u = x - y$ and extend a periodic tail to its translated bi-infinite word. Each exact u level is visited finitely often because u advances $2m$ per period, and every longitudinal cut has algebraic net crossing one. The same local maxima/minima count now gives $\rho_v \equiv \nu_v + 1 \pmod 2$, where ρ_v counts reversals and ν_v visits at $u = v$. L reverses u on horizontal cells and R reverses it on vertical cells. Therefore every horizontal longitudinal line has an odd number of R turns, and every vertical one an odd number of L turns over the bi-infinite trace. Each exact line has finitely many visits, despite the bi-infinite trace. No cell count sums infinitely many period copies.

At primitive diagonal drift, even occupied levels have at least four events unless they are wake levels, by the mod-four law. There is also a sharp second-line restriction. Normalise the lower horizontal rail to level one. On a fixed level $t = x + y$, use x as longitudinal coordinate, so $u = 2x - t$; let the lower rail's single event have coordinate X. Its predecessor on level two is at $X + 1$, and its successor at X. Level-two stable order uses $\tau = i - Nx$, so predecessor precedes successor by $N + 2$ in this order.

To count their ranks, let A_0, A_1 be arrivals from below at even and odd stable rank on a vertical level with m_c events. Counting upward departures and using $m_c = f_{c-1} + f_c$ gives $A_0 = \frac{1}{2}\lceil m_c/2 \rceil$. The singleton rail allows exactly one downward and one upward passage, so there are two nonreversing events. On a vertical level that number is $\lceil m_c/2 \rceil$, hence $m_c = 3$ or 4. There is just one arrival from below, at even rank two, while its downward predecessor has rank zero. The below-arrival indicator is therefore $(0, 0, 1)$ or $(0, 0, 1, 0)$; all remaining events are transverse local minima. Chronology fixes these ranks without imposing a cutoff on row length.

Theorem 5.3 (Minimum diagonal width)**.** *Every finite-support diagonal highway has $W \geq 10$.*

Proof. Width zero cannot support a move, and odd widths are excluded by Theorem 5.1. At width two, successive visits to the middle line are separated by rigid two-step excursions through an extreme. Writing $e = (1, -1)$, upper excursions advance one middle-line site and lower excursions retreat one. Each extreme cell is visited once, so each gap between consecutive sites is crossed at most once in each direction. Positive drift forces net crossing one at every sufficiently far-ahead gap, hence exactly one forward crossing and no backward one. Every sufficiently future middle site is therefore visited once, arriving south and departing east. Its unique turn is L, violating finite support.

For width four, record the destination transverse labels on each future vertical column boundary. Crossings alternate right, left, . . . , right. Rightward labels are $2, 4$ and leftward labels $0, 2$; the extreme labels 4 and 0 may each occur once. Simulating the turn rule in an all-white five-cell column gives exactly these twelve boundary-pair edges:

left word	right words
202	4
4220222	2, 202, 20222, 2022224
2042222	2, 202, 20222, 2022224
202242222	422, 42202, 4220222

Data: A. Jillhewar, exact width-four column classifier, 2026. Table typeset in LaTeX.

The column state records five colour bits and the entry pose. Four flags track extreme crossings at its two boundaries. Exhaustion reaches 22 entry states to depth at most eight, without a state cycle. Assign ranks $3, 2, 1, 0$ respectively to 202242222, the two seven-symbol left words, 202, and all other vertices. Every edge strictly decreases rank. Infinitely many untouched columns would require an arbitrarily long graph path; no path has four edges. This excludes width four. A separate set-based checker reproduces the table. Lean verifies the graph-rank argument, leaving the dynamic extraction unformalised.

Widths six and eight have recurrent one-column behaviour, so acyclicity does not extend. Instead choose respectively six or eight consecutive untouched columns. Retain all 42 or 72 colour bits, the 12 or 16 extreme once-visit flags, and the inward boundary pose. Simulate literal in-block dynamics; after an exit, allow every geometrically possible return through that same outer boundary and forget exterior history. This enlarges the reachable set. The complete relaxed searches reach 357 and 20,320 boundary states respectively, with 555 and 42,004 return transitions. Neither search reaches a right-exit terminal. Positive controls with five and seven columns have 46 and 1,188 right-exit terminals. Separate bitset and cell-set implementations

reproduce the released counts and rejection records.

To apply these certificates, choose the block beyond the finite initial black support and past footprint. Since $x_{qN+i} = qm + x_i$, a genuine highway must enter from the left and eventually exit right. An exterior excursion cannot change longitudinal sides without crossing the block, so its actual return is among the relaxed same-boundary returns. Extreme singleton classes justify the crossing-history restriction. The missing right exit contradicts positive drift. These two computer-assisted reachability results apply at every period and drift magnitude, without a numerical cutoff. □

6 Excluding every growth-four drift

The density bound and heading parity reduce $g = 4$ to three quarter-turn drift orbits: $(2, 0)$, $(2, -2)$ and $(1, -1)$. The first two admit analytic stable-rank contradictions. Here ranks always start at zero and use the exact physical chronology from Theorem 2.1.

6.1 Axis drift: the rank-sign contradiction

For $d = (2, 0)$ put $N = 2M$ and choose phase zero at a vertical arrival at the origin. Horizontal displacement two makes M even. The stable rank uses $\tau_i = i - N\lfloor x_i/2 \rfloor$ within each class $(x_i \bmod 2, y_i)$. Write r_i for phase i's zero-based stable rank. Alternating colours make every L rank odd, whereas the single cyclic tour also determines the rank sum through its permutation sign. Thus $K = \sum_{w_i=L} r_i \equiv \#L \pmod 2$. The single-tour permutation gives the kinematic identity

$$K \equiv 1 + N/4 \pmod 2.$$

The derivation uses the port orders at each quotient vertex. On each quotient row there is one vertical-arrival class Q_y and one horizontal-arrival class C_y. At a class of multiplicity n, let α_r, β_r indicate positive incoming and outgoing ports: N/E at Q, and E/N at C. Order vertices Q_y, C_y by increasing y and put negative ports before positive ones. If $A = \sum_r \alpha_r$, the binary-list inversion count is

$$I(\alpha) \equiv A(n-1) + \sum_r r\alpha_r + \binom{A}{2} \pmod 2.$$

L means unequal port bits at Q and equal bits at C. Summing the two port-list formulae cancels each positive-edge binomial at its source and target. If h_y counts horizontal segments and n_y north segments through the next cut, closed vertical flow gives $h_y = n_{y-1} + n_y$, and the remaining linear terms cancel. Hence

$$K \equiv \sum_v [I(\alpha_v) + I(\beta_v)] + \sum_y \binom{h_y}{2}.$$

Sort outgoing ports and apply the edge map, then undo the incoming-port sort. This factors the successor permutation. Within a directed edge block stable order is preserved, including the cyclic seam. The successor is one even-length cycle, of odd sign. Ordering blocks by row and negative before positive port, its remaining inversion parity E satisfies

$$E \equiv \sum_y n_y + \sum_y n_y n_{y+1}, \qquad \sum_y \binom{h_y}{2} \equiv \sum_y n_y n_{y+1} \pmod 2.$$

Thus $K \equiv 1 + \sum_y n_y = 1 + M/2$, as asserted.

To see the edge-block sign directly, outgoing blocks on row y are $[W_y, E_y, S_y, N_y]$ and incoming blocks $[S_{y+1}, N_{y-1}, W_y, E_y]$. If s_y counts south segments, the reversed pairs give

$$E \equiv \sum_y h_y(s_y + s_{y+1}) + n_y(s_{y+1} + s_{y+2}) \pmod 2.$$

Cut balance $s_{y+1} = n_y$ yields the formula above. For the local correction, expand $\binom{n_{y-1}+n_y}{2} = \binom{n_{y-1}}{2} + \binom{n_y}{2} + n_{y-1}n_y$; the binomial terms cancel in the sum. Finally $M = \sum_y h_y = 2\sum_y n_y$.

The wake theorem places two odd classes in each x residue. An exact row has at most two odd classes, and its signed sum must be zero; its wakes therefore come in opposite-checker pairs. There are two wake rows, with two even-checker odd classes in total. Pairing each horizontal step $u_j = \pm 1$ with its incoming vertical direction $v_j = \pm 1$, their even-phase turn sum is two. Consequently

$$M + 2 + 0 + 2 = \sum_j (1 + u_j)(1 + v_j) \equiv 0 \pmod 4.$$

So $M \equiv 0 \pmod 4$ and $N \equiv 0 \pmod 8$. The rank identity gives $K \equiv 1$, whereas $K \equiv \#L = (N-4)/2 \equiv 0$, a contradiction.

6.2 Even diagonal: the two-lane calculation

For $d = (2, -2)$ choose phase zero at a vertical arrival at the origin, so Q layers are even and C layers odd. Use classes (t, r) with $t = x + y$, $r = x \bmod 2$, and lifted time $\tau_i = i - N(x_i - r)/2$. The same port-sorting factorisation gives $K \equiv 1 + E + \Lambda \pmod 2$. Order edge blocks lexicographically by (t, r, port), with negative port first. Put $D_v = 1$ for an odd class and zero otherwise, let m_v be its number of phase visits, and put $p_v = \sum \alpha + \sum \beta \pmod 2$. Port counting gives $D_Q \equiv m_v - 2p_v$ and $D_C \equiv 2p_v - m_v \pmod 4$. Substituting these into the binary inversion formula, the local correction is $p_v(D_v + 1)$ at a vertical-arrival class and $D_v(p_v + 1)$ at a horizontal-arrival class; Λ is their sum.

Assume all odd classes have one checker parity, as the doubled matching matrices following Theorem 3.2 force at $g = 4$. To compute $E + \Lambda$, write w, e, s, n for directed-block count parities. At even Q layers and odd C layers, flow gives

$$w_{t,r} + e_{t,r} = s_{t+1,r} + n_{t-1,r}, \quad s_{j,r} + n_{j,r} = w_{j+1,1-r} + e_{j-1,1-r}.$$

If wakes are on C, even Q multiplicities imply $w_{t,r} = e_{t,r} = n_{t-1,1-r}$ by finite-support telescoping. All local corrections sum to zero. Each odd $n_{j,r}$ supplies the four blocks $N_{j,r}, S_{j+2,r}, W_{j+1,1-r}, E_{j+1,1-r}$. Their source order is $[N, W, E, S]$ and target order $[W, S, N, E]$, of odd sign. Two gadgets have odd cross-inversion precisely when in the same lane and separated by two layers. Therefore $E = \sum_{j,r} n_{j,r} + \sum_{j,r} n_{j,r}n_{j+2,r}$. This counts runs of ones modulo two. Their boundary is $D_{j,r} = n_{j-2,r} + n_{j,r}$, so if o_r counts lane-r wakes, $E + \Lambda \equiv \sum_{r=0}^{1} \binom{o_r}{2} \pmod 2$. If wakes are on Q, interchange the flow roles: the gadgets are $S_{j,r}, N_{j,r}, E_{j-1,r}, W_{j+1,r}$ with source/target orders $[E, S, N, W]$, $[S, W, E, N]$; the same run calculation gives the formula. A common checker sign forces $o_r \equiv 2 \pmod 4$ in both lanes by Theorem 3.2, so both binomials are odd and $K \equiv 1$. But horizontal displacement two forces $M = N/2$ even, and heading reset makes $g/2$ even; hence $K \equiv \#L = M - g/2 \equiv 0$. Hence every highway at this drift has odd classes on both

checker parities, excluding the forced-pure growth-four case.

6.3 Primitive diagonal: a symbolic certificate without cutoffs

At $d = (1, -1)$ each transverse level is one class. Proposition 5.2 and extremal rigidity force three horizontal wake levels and one vertical wake level, with $N \equiv 6 \pmod 8$. Nesting the top-rooted transverse excursions gives a finite ordered tree: each child is a successive inward excursion along the contour. Its count and stable-order laws force a prefix shared by every candidate; excluding that prefix excludes the whole period. At inward depth h, a node's longitudinal voltage is $v = (-1)^{h+1} + \sum_z v_z$, summed over its ordered children. For a row with n nodes, m children, p nonleaves, and $\omega = (n + m) \bmod 2$, letter counting gives

$$4p = n + m + \omega \quad (h \text{ odd}), \qquad 4p = n + m - \omega \quad (h \text{ even}).$$

Stable order is decreasing longitudinal coordinate with contour order breaking ties. A two-child top branch leaves an empty integer L gap and is impossible. For three children, the next count is $1, 2, 5, 6, 9$ or 10; leaf equations remove $1, 2$ and endpoint gaps remove $9, 10$. The remaining cuts are $(3, 5)$ and $(3, 6)$ with node bases $(-1, -2, -1)$ and voltages $(-1, 1, 0)$.

Their four fixed L anchors lie at $0, 0, -1, -2$. The singleton leaf R at -1 fills one gap. The other gaps force interior coordinates $0, -1$, a leading coordinate α, and, for six children, a trailing $\beta \le -2$. Permuting these forced entries gives eight affine five-child forms and forty six-child forms; $\alpha \ge 0$ or $\alpha \ge 1$ according to the node containing it. The row-count law leaves eight possible next cuts from five children and nine from six. For the five-child cuts and six-child cuts $2, 5, 6$, the verifier solves leaf equations and partitions integer parameters at every endpoint-order and unit-gap crossing, including equality cells and unbounded tails. It permits omitted L events in any remaining integer gap, a relaxation of the exact contour. No candidate survives this relaxed test. The larger six-child cuts use the endpoint obstructions described below.

The verifier tests the displayed families; the affine-form reduction must establish coverage. Let q and r be the two nonleaf coordinate strings beginning at $-2, -1$ and ending at zero, and let π range over all permutations. The five-child forms are

$$\begin{aligned} &q = (-2, 0), && r = (-1, \pi(\alpha, 0, -1), 0), && \alpha \ge 1, \\ &q = (-2, \alpha, 0), && r = (-1, \pi(0, -1), 0), && \alpha \ge 0. \end{aligned}$$

They number $6 + 2 = 8$. The six-child forms are

$$\begin{aligned} &q = (-2, 0), && r = (-1, \pi(\alpha, 0, -1, \beta), 0), \\ &q = (-2, \alpha, 0), && r = (-1, \pi(0, -1, \beta), 0), \\ &q = (-2, \beta, 0), && r = (-1, \pi(\alpha, 0, -1), 0), \\ &q = (-2, \pi(\alpha, \beta), 0), && r = (-1, \pi(0, -1), 0). \end{aligned}$$

Here $\beta \le -2$; $\alpha \ge 1$ in the first and third rows and $\alpha \ge 0$ in the others. They number $24 + 6 + 6 + 4 = 40$. Successive coordinate differences determine the child voltages, so these are affine parameter families rather than sampled integer values. For five children the next cuts are $2, 3, 6, 7, 10, 11, 14, 15$; for six they are $2, 5, 6, 9, 10, 13, 14, 17, 18$. For the six-child cuts $9, 10, 13, 14, 17, 18$, obstruction U1 forces the last node to be a leaf: otherwise its final R and the first node's R leave no integer L gap. U2 forces a leaf at the two-block junction: otherwise its adjacent R anchors coincide. At cuts $9, 10$, the two forced leaves give consecutive L

events; at 13, 14 only one leaf is available, and at 17, 18 none. Thus these cuts also fail alternation. The count and affine-form reduction is documented in `primitive_g4_cap5_symbolic_2026-08-22.md` [10]. The JSON record summarizes counts, cuts, endpoint obstructions and a checker hash. The checker regenerates the parameter cells and degree compositions. The JSON records the results, rather than a full proof object.

This computer-assisted classification has no period cutoff. Completeness depends on the affine-form reduction and integer-cell verifier. An independent audit regenerated the eight and forty inputs and reran the verifier, without a second complete symbolic implementation. The classification is not fully formalised in Lean. The finite premise is recorded in `primitive_g4_top_prefix_certificate.json` and checked by `audit_primitive_g4_top_prefix.py` in the companion archive. Theorem 5.3 supplies the inward rows required by this primitive-prefix test. Combining all three drift exclusions proves

$$\boxed{g \in 4\mathbb{Z}_{\geq 2}, \qquad g \geq 8.}$$

7 Selected cycles, detour charges and period bounds

Choose a representative c_v of each visited translation class. Lifted times at c_v are $i - Na_i$. Let F_v, T_v be the first and last such times. An edge from c_v to $c_u + \lambda d$ has integer voltage λ.

Theorem 7.1 (Cycle and detour bounds)**.** *The forward first-entry cycle has voltage one. Its length k and wait lengths D_i satisfy*

$$N = k + \sum_i D_i, \quad D_i \in 4\mathbb{Z}_{\geq 0}, \quad k \geq 2\|d\|_\infty.$$

Its nonzero waits have integer charges q_i with $g = 4\sum_i q_i$, and

$$\boxed{N \geq 6\|d\|_\infty + 2\|d\|_1.}$$

Proof. A selected last-exit edge satisfies $T_u + N\lambda \geq T_v + 1$. Summing around a functional-graph cycle gives positive voltage; reversed first-entry edges similarly give negative voltage. Such a cycle has distinct quotient vertices, no perpendicular-axis backtrack, and is an embedded circle on the translation cylinder. An essential embedded circle winds once, so its voltage is $+1$ or -1. Reverse the first-entry cycle to orient it forward.

Lift it sufficiently far ahead. Between first arrival at a source and the departure feeding the next first arrival, the actual trace makes a closed physical wait. Together, the waits and selected edges lead to the next translated first visit, exactly N steps later. Closed quarter-turn walks have even horizontal and vertical step counts, hence lengths divisible by four. The selected cycle alternates axes, giving its bound on k. It has equal numbers of geometric right and left corners: an essential cylinder circle has total signed turning zero. A geometric right corner has zero wait because its white first visit immediately takes the selected edge. A left corner requires a positive wait.

At a positive wait let T_i be its turn sum excluding the selected departure and $s_i = \pm 1$ that departure's turn. If the terminal incoming heading is unchanged, $T_i \equiv 0$ and $s_i = -1$; if reversed, $T_i \equiv 2$ and $s_i = 1$. Thus $q_i = (T_i + s_i + 1)/4$ is integral. Pairing each positive wait with a zero-wait right corner gives $g = 4\sum_i q_i$.

A four-step positive wait is a clockwise unit square. Its third step turns R at the predecessor of its root's first arrival, so that predecessor's earlier selected departure turned L. Such a wait cannot follow a geometric right corner. If ρ counts cyclic right-to-left corner transitions, all $k/2$ nonzero waits cost at least four steps,

and these ρ waits cost at least eight. Hence $N \geq 3k + 4\rho$. For selected unit steps e_i and corner signs $c_i = \pm 1$, $e_{i+1} = -c_i c_{i+1} e_{i-1}$ gives $d = \sum_{c_i \neq c_{i+1}} e_{i-1}$. There are 2ρ changes, so $\|d\|_1 \leq 2\rho$. Combining the inequalities proves the result. □

For a simple positive wait the root's closing corner is right. If its polygon were counterclockwise, that corner would be reflex and the feeding edge would enter its bounded interior. The future first-entry ray cannot meet the already visited polygon, so it could not escape to infinity. Thus the polygon is clockwise, has turn sum four, and charge one. General waits need not have positive charge. The locally valid eight-step word RRRLLLLR can return with turn sum zero and feed a left departure, giving $q = 0$. This local detour does not realise a highway. The charge identity gives no upper bound on period or excursion complexity.

7.1 Why short detours cannot account for positive growth

Proposition 7.2 (Eight-step waits)**.** *Every positive first-entry wait of length eight has charge zero.*

Proof. An intermediate return to its root could occur only at step four, since closed quarter-turn walks have lengths divisible by four. If that first square is clockwise, the black root returns with its initial heading; its L departure already takes the selected edge, reaching the next selected first visit too early.

For the other orientation, normalise the initial heading to N. The square has directions E,N,W,S and turns R,L,L,L. At step four the black root has heading S, and its L turn again sends it E. That next cell was flipped black to white in the first square, forcing R and S. Closing the second square then requires W,N. At step eight the root is white with heading N, so its R departure is E, whereas the selected geometric left edge is W. This also cannot feed the selected edge. Thus the root has no intermediate visit.

Its final colour is consequently black. The feeding turn is L, its incoming heading is unchanged, and the wait's turn sum T is divisible by four. There is no simple eight-edge lattice polygon that turns at every vertex: its four horizontal and four vertical edges confine its bounding box to a two-by-two square. Its enclosed faces are edge-connected; perimeter eight leaves three faces in an L or all four faces, and both boundaries have consecutive collinear edges.

Another cell must therefore repeat. Its visits are four steps apart, and splitting the cyclic walk there gives two unit squares. Let their clockwise orientation signs be $\epsilon_1, \epsilon_2 \in \{1, -1\}$. Their six other corners contribute $3(\epsilon_1 + \epsilon_2)$. The two actual turns at the repeated cell alternate and cancel. The root's closing corner equals its initial R corner because its heading returns unchanged. Hence $T \in \{-6, 0, 6\}$. Only zero is divisible by four, giving $q = (T - 1 + 1)/4 = 0$. □

Proposition 7.3 (Coupled predecessor restriction)**.** *At a cyclic geometric R,R,L pattern of a selected first-entry cycle, the wait at the L corner has length at least twelve.*

Proof. The preceding R already rules out length four. For length eight, the two unit squares above have opposite orientations. Coincident squares would force a reversed join, and squares meeting only at opposite corners would force a straight join. Both violate the quarter-turn rule. They share an edge; the root, which does not repeat before the end, is a nonshared corner of the clockwise square.

There are only two such corners. With root $v = (0, 0)$ and initial heading E, their direction words are

$$A: \quad S, W, N, W, S, E, N, E,$$
$$B: \quad S, W, S, E, N, W, N, E.$$

The selected feeding edge is north. Set the first visit to v at time zero. Its cycle predecessor $u = (-1, 0)$ departs at time -1. In case B, the next visit to u, at time seven, turns R; alternation forces its selected departure to have turned L. Therefore u cannot be a geometric R corner, whose selected departure is its first R.

In case A, u is visited at times three and seven with turns L and R. If u is a geometric R corner, its first arrival at time -1 has heading N, so its predecessor is $z = (-1, -1)$, departing north at time -2. The wait reaches z at time two and turns R. The intervening positions $u, v, (0, -1)$ avoid z, so its selected departure must have turned L. Thus the corner before u cannot also be R. This excludes R,R,L in either case, including across the period seam. □

Corollary 7.4 (A necessary long wait)**.** *Every selected first-entry cycle of a positive-growth highway contains a wait of length at least twelve.*

Proof. Suppose all nonzero waits had length four or eight. A run of two R corners would end in R,R,L and force a longer wait. Hence every R run is a singleton. The cycle has equally many R and L corners and equally many runs of each, so every L run is also a singleton. Its turns alternate R,L. Every positive wait follows R, excluding length four; all have length eight and charge zero. The exact charge identity would give $g = 0$, a contradiction. □

Longer waits still have no charge-sign bound. The argument neither bounds period from above nor excludes growth eight.

Corollary 7.5 (Strict diagonal period bound)**.** *For drift $d = (m, -m)$, $m \geq 1$, every highway has $N \geq 10m+4$.*

Proof. Equality in $N \geq 3k + 4\rho \geq 10m$ forces $k = 2m$ and $\rho = m$: corners alternate and all nonzero waits have length eight. Their zero charges would give $g = 0$, contradicting positive growth. Thus $N > 10m$; the congruence $N \equiv 2m \pmod 4$ gives the stated bound. □

8 Further exact structural results

Physical returns and black-set topology give further identities. The arithmetic tests also show where static information fails to determine a trajectory.

8.1 Tait routing and exact boundary surgery

In planar routing, a cell flip splits or merges circuit boundaries. The black set's topology describes this operation exactly. Normalise even cells to have vertical arrivals. Under the frozen white routing, directed arcs form clockwise four-cycles around one checkerboard class of plaquettes. Label those plaquettes by their

lower-left corners. An ant cell $z = (x, y)$ corresponds to the Tait edge

$$e_z = \begin{cases} \{(x, y-1), (x-1, y)\}, & x + y \text{ even}, \\ \{(x-1, y-1), (x, y)\}, & x + y \text{ odd}. \end{cases}$$

The black cells give a finite edge subgraph H_B. White and black select the two smoothings at a medial crossing: with its four arc ends cyclically labelled, they connect 1–2, 3–4 and 1–4, 2–3, respectively. Thickening Tait vertices to discs and occupied edges to bands shows that frozen routing circuits are exactly boundary circles of a regular neighbourhood of H_B, together with untouched white four-cycles. The live ant follows the selected local arc and then switches that smoothing. Switching two arcs on one circle splits it; switching arcs on two circles merges them. This proves the boundary-surgery description directly from the turn rule.

For E edges, V incident vertices, k nonempty components and $\beta = E - V + k$, the neighbourhood has Euler characteristic $V - E$. It consists of k planar surfaces, so if C counts its boundary circles, $V - E = 2k - C$. Relative to the V separate white loops, $\Delta C = E - 2V + 2k, \qquad E + \Delta C = 2\beta$. This counts finitely many affected loops relative to the infinite white plane. Inserting a bridge merges two loops and leaves cycle rank fixed. Inserting a cycle edge splits a loop and raises cycle rank one; deletion reverses these operations. A sufficiently advanced bounding-box record beyond the initial support creates a white edge with a previously untouched Tait endpoint, hence a bridge. Cycle-rank creation can still occur inside explored territory.

Proposition 8.1 (No affine monotone graph statistic)**.** *No affine combination of (E, V, k, β) that is nonconstant on the allowed graph statistics is monotone even along the blank orbit.*

Proof. Since $\beta = E - V + k$, reduce such a combination to $\lambda \cdot (E, V, k) + c$ with $\lambda \neq 0$. Inserting an edge has exactly four cases: two new endpoints, one new endpoint, two old endpoints in distinct components, or two in the same component. Their increments are $(1, 2, 1), \quad (1, 1, 0), \quad (1, 0, -1), \quad (1, 0, 0)$. Deletion supplies their negatives. The archived blank-orbit witness checker reproduces all eight within its first 12,000 steps. These finite occurrences suffice, regardless of asymptotic frequency. The set is symmetric and spans $\mathbb{R}^3$. For every nonzero λ some increment has positive dot product and its realised negative has negative dot product. The statistic therefore both increases and decreases on that orbit. □

Distant connectivity can change a local smoothing's merge/split status. The topological identities therefore leave information that a bounded visible window cannot determine.

8.2 Incidence parity and physical black lifetimes

Map (x, y) to an edge $C_x R_y$ of the simple bipartite column–row graph and work over $\mathbb{F}_2$. Let $\eta_t = C_{p_{t,x}}$ at a vertical heading and $R_{p_{t,y}}$ at a horizontal heading. A toggle has boundary $C_x + R_y$; turning changes token type, and movement preserves the coordinate of the new token. Telescoping gives the exact identity $\partial(B_s \triangle B_t) = \eta_s + \eta_t$. The discrepancy has zero or two odd vertices; in the latter case they lie in one component. All other components are Eulerian. For consecutive R and L visits at a cell z at times $r < l$, put $E = B_{r+1} \triangle B_l$. The cell z is absent from E, but $\partial E = C_{z_x} + R_{z_y}$. The path connecting those vertices cannot be the unavailable edge z and has at least three edges. Consequently $|B_r \triangle B_l| \geq 4$: at least three other cells

have odd visit count. The interval runs from just after R to just before L. Different directed poses remain distinct, and the calculation gives no return-time upper bound.

Proposition 8.2 (A six-cell physical return bound)**.** *Suppose an interval returns to the same cell z with opposite incoming heading and with z unchanged in colour. At least six physical cells change colour over that interval.*

Proof. Write $D = \mathbf{1}_{B_t} - \mathbf{1}_{B_s}$ for the signed black-set difference and G for its support, the symmetric difference. The unchanged root means $z \notin G$. The endpoint tokens have the same axis and coordinate, so $\partial G = 0$: its simple bipartite incidence graph is Eulerian. Use the mod-four charge with weight $\chi(x, y)\mathbf{1}_{x=z_x}$. The explicit potential table has $F_{q+2}(z) - F_q(z) \equiv 2w(z)$ for every heading. Conservation therefore makes the signed checker sum in column z_x equal to two modulo four. The coordinate-swapped charge gives the same result on row z_y. In particular both root fibres contain changed cells.

A nonempty simple bipartite Eulerian graph has at least four edges, and its edge count is even, by summing even degrees on one side. If it had four, it would be a single rectangle. Since it uses the root row and column, one of its four edges would be their intersection z, contrary to $z \notin G$. The next possible size is at least six. □

These six physical cells and their signs can merge under translation projection. Their count cannot be added directly to a wake-class budget.

8.3 Black-count geometry and polynomial charges

Put $b_t = |B_t|$ and $J_t = \sum_{z \in B_t} z - b_t p_t$. Lemma 1.1 gives

$$p_{t+1} - p_t = u_{q_0 - b_0 + b_{t+1}}, \qquad J_{t+1} - J_t = -b_{t+1} u_{q_0 - b_0 + b_{t+1}}.$$

For the second formula, toggling adds $r_t p_t$ to the first moment, which cancels the change of the centred term at the old position. Thus a nonnegative count walk determines a planar path, but that path still has to pass physical colour consistency. Only a clean translation forces $J_N = J_0$; ordinary growing highways need not satisfy it.

The four-corner condition classifies affine mod-four weights: $w = \alpha x + \beta y + \gamma$ is admissible exactly when $\alpha \equiv \beta \pmod 2$. For a quadratic $w = ax^2 + bxy + cy^2 + dx + ey + f$, coefficient comparison gives $b \equiv 0 \pmod 2$, $\quad 2a + b + 2c + 2d + 2e \equiv 0 \pmod 4$. Modulo affine, constant and doubled row/column charges, the new class is $x^2 + 2xy$, of order two. Modulo eight the torsion and corner conditions force every coefficient even and b divisible by four; dividing by two recovers the mod-four family. There is no primitive quadratic lift to modulo eight. This classification covers affine and quadratic additive charges; these charges alone do not decide realisability.

For the affine representative $w = x + y$, the corrections are

$$(F_N, F_E, F_S, F_W) = (2x + 3y + 3,\ x + 2y + 2,\ y + 3,\ 3x + 2).$$

For $Q = x^2 + 2xy$, they are

$$F_N = 2xy + 3x, \quad F_E = 3x^2 + x, \quad F_S = 2x^2 + 2xy + 3x, \quad F_W = x^2 + x.$$

Substitution into the eight local equations verifies both tables. To obtain the quadratic conditions, expand the four-corner sum: the variable terms reduce modulo four to $2b(x + y)$, and its constant term is $2a + b + 2c + 2d + 2e$. Their vanishing is equivalent to the displayed coefficient constraints. Even square coefficients are doubled row/column charges; $y^2 + 2xy$ differs from Q by the old charge family, and $2Q = 2x \pmod 4$. Thus the remaining quotient class has order two.

The modulo-eight conclusion also has a coefficient proof. First $4w = 0$ forces $a \equiv d$, $c \equiv e \pmod 2$ and b, f even. The variable terms in the corner equation give $b/2 \equiv d \equiv e \pmod 2$. Its constant part eliminates the odd choice. Hence a, c, d, e, f are even and b is divisible by four. Dividing the equations by two gives exactly the admissible mod-four quadratic family. A primitive quadratic lift is therefore impossible. Higher-degree and non-additive invariants remain outside this argument.

8.4 Exact count spectra and an infinite compatible family

The centred-moment identity also gives an exact arithmetic model of a prospective clean translator. Consider a closed nonnegative integer walk b_t with increments $+1, -1$, interpreted formally as R,L. Put $e_k = u_{q_0 - b_0 + k}$, so a step arriving at height k has direction e_k, with the four directions repeating clockwise. Let m_k count upward crossings of $k \leftrightarrow k + 1$. Closure supplies the same number of downward crossings. Thus

$$N = 2\sum_k m_k, \qquad d = \sum_k m_k(e_k + e_{k+1}), \qquad Z = \sum_k m_k\big(ke_k + (k+1)e_{k+1}\big).$$

Here the centred-moment change is $-Z$; clean translation would require $Z = 0$. These formulas depend only on crossing multiplicities, not on their Euler ordering. They retain absolute black-count heights: subtracting the minimum height generally changes Z by a multiple of the drift and is therefore not a harmless normalisation.

Rotate axes so $e_0 = (1, 0)$ and let $H = \sum_k m_k$. The four values of $v_k = ke_k + (k + 1)e_{k+1}$, according to k mod 4, are $(k, -k - 1)$, $(-k - 1, -k)$, $(-k, k + 1)$, $(k + 1, k)$. Their second coordinates are all -1 modulo four. Consequently $Z = 0$ forces $H \equiv 0 \pmod 4$ and $N \equiv 0 \pmod 8$. This restriction follows within the formal count model. The associated path must still satisfy physical colour consistency.

Proposition 8.3 (An unbounded zero-moment count family)**.** *For every integer $h \geq 3$, there is a closed nonnegative count walk of length $8h$ with nonzero displacement and $Z = 0$.*

Proof. Put $a = h - 2$ and prescribe the four consecutive multiplicities $(m_a, m_{a+1}, m_{a+2}, m_{a+3}) = (h, h + 2, h - 2, h)$. In axes rotated so that $e_a = (1, 0)$, their displacement is $(0, -4)$. Direct substitution gives $Z = (0,\ 4(h - a - 2)) = (0, 0)$, $\qquad N = 2(4h) = 8h$. Every multiplicity is positive. Replace each height edge by that many arcs in each direction. The resulting finite directed multigraph is connected and balanced, so an Euler circuit starting at a exists. It supplies the required closed walk without ever falling below the nonnegative height a. □

These zero-growth spectra satisfy exact moment cancellation at arbitrarily large periods, yet cannot be finite-support ant translators by the clean-translation obstruction: repeated physical cells must still alternate.

The unbounded minimum height can nevertheless be handled exactly at a fixed period. Traversed height edges form a contiguous interval, so their positive multiplicities are an ordered composition of $H = N/2$.

There are 2^{H-1} compositions. Write its lowest height as $a = r + 4s$, where $0 \le r < 4$ and $s \ge 0$. For a fixed composition and residue, let d_r, Z_r denote the displayed sums with minimum height r. Periodicity of e_k gives $d(a) = d_r, \qquad Z(a) = Z_r + 4s\, d_r$. If $d_r \neq 0$, at most one integer $s \ge 0$ can make the vector $Z(a)$ vanish. Checking that value for each composition and each of four residues therefore covers every nonzero-drift moment-compatible count spectrum without a height cutoff. The reduction enumerates multiplicities; Euler orders and physical trajectories still require separate checks [10]. For a prescribed nonzero drift, compositions with the wrong d_r are discarded first. The remaining scalar candidate for s must be a nonnegative integer satisfying both coordinate equations; solving only one component would admit false moment matches.

8.5 Voltage-aware lifetime rotation and cycle spaces

Translated event times let the incidence argument track black lifetimes across period boundaries. For each stable consecutive R/L pair, write its representative cell as c_e and its lifted times as $r_e < l_e$. Translating by nd adds nN to both times. Far enough ahead, these are actual black lifetimes at one physical cell. Every L belongs to one such pair; there are $m = (N - g)/2$ pair types and g unmatched R wake types.

Put $\lambda_e = l_e - r_e$. The least seed at a pre-turn cut comprises exactly the paired lifetimes awaiting L, with disjoint occupied roots:

$$\mu_s = \sum_e \sum_{n\in\mathbb{Z}} \mathbf{1}_{\{r_e+nN<s\le l_e+nN\}}, \qquad \sum_{s=0}^{N-1} \mu_s = \sum_e \lambda_e.$$

The map $(s, n) \mapsto s - nN$ counts the λ_e integers in $(r_e, l_e]$ once; terminal unmatched R events contribute no seed. Each closed return alternates axes with even move counts on each, so $\lambda_e \in 4\mathbb{Z}_{>0}$. Set $h = \gcd(N, 4)$ and $\varepsilon_i = 1$ for R, -1 for L. Paired phases agree modulo h and cancel, leaving $\sum_{0\le i<N,\ i\equiv j\ (h)} \varepsilon_i \ge 0$ for every residue j: a necessary word-only acceptance condition.

Let $A_{ef,n} = 1$ when precisely one endpoint of the translated pair $f + n$ lies strictly inside (r_e, l_e), and zero otherwise. Let $U_{eu,n}$ indicate an unmatched wake copy inside that interval. The odd-visit vector on $[r_e, l_e)$ is, over $\mathbb{F}_2$,

$$v_e = \delta_{c_e} + \sum_{f,n} A_{ef,n}\delta_{c_f+nd} + \sum_{u,n} U_{eu,n}\delta_{c_u+nd}, \qquad \partial v_e = 0.$$

The first term is its initial R visit. A pair contributes precisely when one endpoint is inside; two endpoints cancel. The half-open walk starts and ends at the same cell and alternates axes, so the numbers of visits on each exact row and column are even, proving its zero boundary. All sums are finite. Exchanging two interlaced intervals and shifting representatives by $-n$ gives $A_{ef}(T) = A_{fe}(T^{-1})$, where $A_{ef}(T) = \sum_n A_{ef,n}T^n$. Diagonal Laurent terms can be nonzero: a long lifetime may interlace translated copies of its own type.

Orient the same coefficient by setting $K_{ef,n} = 1$ if only the other R endpoint is inside, -1 if only its L endpoint is inside, and zero otherwise. The two crossing interval orders give opposite signs, while nested or disjoint intervals give zero in both directions. Thus $K_{ef,n} = -K_{fe,-n}$ over the integers. Close lifetime e at its root. Its closing quarter-turn is $\eta_e = 1$ for parallel incoming headings at R and L, and -1 for opposite headings. Every interior turn is the signed contribution of another paired endpoint or an unmatched R. If ω_e

is the integer rotation number with R positive, the closed-walk turn sum gives

$$\eta_e + \sum_{f,n} K_{ef,n} + \sum_{u,n} U_{eu,n} = 4\omega_e.$$

Sum over pair types. Skew reciprocity cancels all paired terms, yielding the exact global identity

$$\boxed{m - 2E_{\mathrm{opp}} + H_{\mathrm{tot}} = 4\sum_{e} \omega_e.}$$

Here E_{opp} counts opposite-arrival pairs and H_{tot} counts contained unmatched event copies, rather than distinct wake types. Self-intersections permit negative, zero and larger positive rotation numbers. For the displayed standard highway, $m = 46$, $E_{\mathrm{opp}} = 22$, $H_{\mathrm{tot}} = 190$, and $\sum_e \omega_e = 48$, giving $46 - 44 + 190 = 4 \cdot 48$.

The lifetime pairs also define a labelled graph. Identify the paired phase occurrences in the temporal circuit to obtain four-valent lifetime vertices; unmatched wakes are two-valent H/V switches. Smooth each paired vertex into its H/H and V/V strands. Cutting the wake switches leaves $g/2$ open H trails, $g/2$ open V trails and c closed circuits. The boundary touch graph has smoothing components as vertices. Its edges carry lifetime labels or mark wakes. Its wake subgraph is two-regular and bipartite, with κ even cycles.

Suppressing the wake switches produces a connected four-regular temporal graph. Its touch graph T_0 is the boundary touch graph with wake cycles contracted. It has m edges and $c + \kappa$ vertices; hence $\dim Z(T_0; \mathbb{F}_2) = m - c - \kappa + 1$. After deleting wake coordinates, lifetime rows are the modified interlacement rows: the H/H,V/V transition has the required diagonal-one correction, and extra complete period laps visit each pair twice, cancelling modulo two. The circuit-partition theorem then identifies their span with this labelled cycle space [9].

The contraction statement has an explicit cycle-space explanation. Write T_∂ for the labelled boundary touch graph and W for its wake-edge subgraph. Deleting wake-edge coordinates gives an exact sequence over $\mathbb{F}_2$,

$$0 \longrightarrow Z(W) \longrightarrow Z(T_\partial) \longrightarrow Z(T_0) \longrightarrow 0.$$

A quotient cycle lifts by joining its endpoints inside each connected wake component; these joins restore even degree. The kernel consists exactly of cycles supported entirely on W, one independent direction for each of its κ cycles. Spanning the contracted space leaves at most κ missing labelled directions; the lifetime rows need not supply them. The standard example has contracted dimension 41 and full boundary dimension 42; its full lifetime-row rank remains 41, before physical coordinate identification. Physical row and column identification can shrink the labelled lifetime-row span, so equality with the full physical incidence cycle space is unproved. These dimensions and the wake budget do not bound c. Finite-state closure and further growth exclusions remain open.

9 A realised highway attaining width ten

For a north-facing ant at the origin, concatenate these four lines:

RRRRLLRLLRRRRLLRRRRLLRLRRR
RLRLLLLRRRRLRRLRRRRLLLLRLR
RRRLRRRRLLLLRLRRRRLRLLRRLL
LLRRLLRRRRLLRRLRLLRLLRLRLL

The word has $N = 104$, $d = (2, -2)$, $g = 12$ and $W = 10$. Figure 1 shows its geometry.

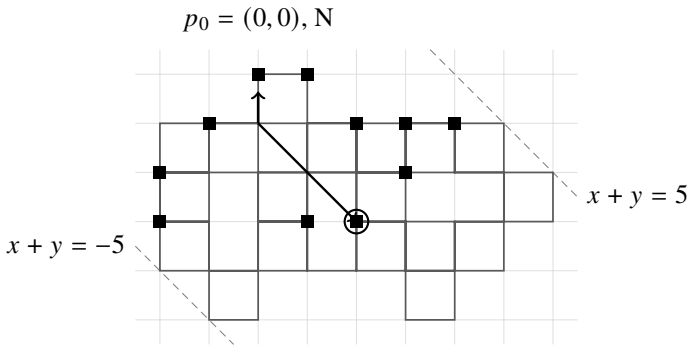


Figure 1: The actual 104 moves and eleven-cell seed. Squares are initially black; repeated trace edges overlap. The long arrow joins the north-facing origin to its outlined endpoint and denotes net drift, not one move. Dashed rails attain labels $x + y = \pm 5$, whose difference is $W = 10$, not their Euclidean separation.

Data: A. Jillhewar, 2026; diagram: algorithmic rendering with Python and LaTeX/TikZ, 2026.

The seed supplied by Theorem 2.1 is

$$\{(-2, -2), (-2, -1), (-1, 0), (0, 1), (1, -2), (1, 1), (2, -2), (2, 0), (3, -1), (3, 0), (4, 0)\}.$$

The criterion and endpoint decomposition prove indefinite repetition and $|B_{104n}| = 11 + 12n$. Simulation checks the sampled period boundaries. This attains Theorem 5.3; minimum growth and highway uniqueness remain open. Its 22 classes are twelve odd and ten even, giving $g = 12$. Each has level range at most two; eleven of eighteen exceptional class–level combinations begin L and supply the seed cells.

The cross-period chronology example above works through one of these classes explicitly. The level-gap formula gives minimum seed size 11 over the 104 cyclic shifts, attained only at shifts $0, 33$ with headings normalised by quarter-turns. This minimum concerns immediate repetition of this word, excluding transients and other highways [10]. Shift by 100 symbols and half-turn the translated pose to face north. This first-L phase has $d = (-2, 2)$ and a thirteen-cell least seed (`normalisation_B` in `standard_highway.json` [10]). Alternative wake bases give two fixed cap cells, eleven translating cells and twelve growing strands: $|B_{104n}| = 13 + 12n$. Its 32 toggled cells occupy 21 signed-field orbits (twelve growing, nine zero-sum), distinct from its 22 visited classes.

10 Exact finite-period searches and verification

The criterion is an exact acceptance test. A phase of least seed size starts R and has nonnegative turn prefixes by the seed-phase recurrence, so this normalisation preserves every realisable candidate. The through-32 baseline fixes only initial R; later engines also enforce nonnegative prefixes. A length-k prefix rank encodes $k - 1$ binary symbols following R, giving 2^{11} ranks for $k = 12$, including pruned prefixes. Accepted leaves

restore heading, have nonzero drift and pass exact coset chronology. Positive-growth pruning uses the proved zero-growth exclusion.

Chronological completion pruning can be proved directly. If the known symbols in a class's eventual stable order are $s_1 \cdots s_m$, the minimum insertions for an alternating completion starting R and ending L are

$$\begin{aligned} \mathrm{need}_R &= \mathbf{1}_{s_1=L} + \#\{i : s_i = s_{i+1} = L\}, \\ \mathrm{need}_L &= \#\{i : s_i = s_{i+1} = R\} + \mathbf{1}_{s_m=R}. \end{aligned}$$

Each indicated endpoint or equal-symbol gap requires a distinct insertion; inserting one opposite symbol in every such gap achieves the bound. Both needs are zero for an empty word. If an odd final class is allowed, omit the terminal-L requirement. Summing needs across classes and comparing with the remaining R/L budget is a sound rejection rule.

Archived residue-theorem-free exhaustive runs exclude every nonzero-drift period $N \le 48$. The period-48 run records 41,710,394,384 nodes and 7,446,719,550 exact evaluations. Growth-eight runs at $50, 52, 54, 56, 58, 60$ cover all 2,048 normalised length-twelve prefix ranks at each period, with zero accepted words. Thus

$$N > 48 \quad \text{for every highway,} \qquad N > 60 \quad \text{if } g = 8.$$

Periods 50–58 use residue pruning; period 60 uses a hybrid endpoint-deficit engine with an audited bounded suffix-reachability calculation. These exclusions depend on complete rank coverage and exact leaf acceptance. The recorded engine versions must also match the runs. A partial zero-hit run proves no complete exclusion; the archived long searches were not rerun.

An independent Python checker agrees through length 18: 524,286 words, including 172,092 heading-reset nonzero-drift words. This checks the criterion on the stated range. Longer search completion requires separate evidence. The Java variants share infrastructure. Independently structured width checkers reproduce local counts; `aggregate_indep.py` audits period-rank completion. The archived manifest identifies checkers, period-60 engines and outputs [10]. Lean verifies conservation/parity kernels and the width-four rank argument. The other dynamic reductions and complete growth-four verifier remain unformalised.

11 The unresolved growth-eight boundary

The first unexcluded growth is eight. The density theorem gives $\|d\|_\infty \le 4$, while heading parity and signed residues leave 40 nonzero drifts in ten quarter-turn orbits. Representatives are

$$(1,-1),\ (2,0),\ (2,-2),\ (1,3),\ (1,-3),\ (3,-3),\ (4,0),\ (2,4),\ (2,-4),\ (4,-4).$$

Wake arithmetic and partial-rail or cap arguments restrict subfamilies. Entire drift orbits remain unexcluded. For $d = (1,-3)$, lifted clocks distinguish physical returns from quotient returns. The archived upper-cap normal form with inward multiplicities $(1,2,2)$ is excluded at every width and period by return-heading or clock contradictions; this does not exclude the lower orientation [10]. All ten orbits retain open cases. Establishing existence requires a word that passes Theorem 2.1; static wakes and relaxed clocks are insufficient. Whether a growth-eight highway exists remains open, as does eventual highway entry from every finite seed. In particular, $g \ge 12$ and universal entry are still unproved.

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